

Cyclistic Bike-Share Analysis Case Study

This case study analyzes Cyclistic bike-share data to understand the behavioral differences between casual riders and annual members. The project was completed using Google BigQuery for SQL analysis and Power BI for data visualization.

1. Ask

Business Task

Cyclistic wants to increase the number of annual memberships. The marketing team believes casual riders can be converted into annual members because members are more profitable for the company.

Objective

Identify behavioral differences between casual riders and members using ride data analysis.

Key Questions

1. How do casual riders and members use bikes differently?
2. Why would casual riders buy memberships?
3. How can digital media encourage membership conversions?

2. Prepare

Dataset

The dataset used for this analysis was obtained from the Divvy bike-share public dataset. The data contains ride information such as ride type, ride duration, rider category, station details, and timestamps.

Tools Used

- Google BigQuery
- SQL
- Power BI

Data Credibility

The dataset is reliable, original, comprehensive, current, and cited (ROCCC framework).

3. Process

Data Cleaning

The datasets from January 2024 to April 2024 were combined using UNION ALL in BigQuery. Ride length and weekday columns were created using SQL functions. Invalid rides where ended_at was earlier than started_at were removed.

SQL Operations

- Combined monthly datasets
- Created ride_length column
- Extracted day_of_week
- Filtered invalid ride durations

4. Analyze

Key Findings

- Members generated significantly more rides than casual riders.
- Casual riders had longer average ride durations.
- Members were more active during weekdays.
- Casual riders showed higher activity during weekends.
- Classic and electric bike usage were relatively balanced.

5. Share

Dashboard

An interactive Power BI dashboard was created to visualize:

- Total rides by rider type
- Average ride length
- Riding trends by weekday
- Bike type preference

6. Act

Recommendations

- Offer weekend promotions to convert casual riders into members.
- Introduce commuter-focused membership benefits.
- Increase marketing campaigns targeting recreational riders.
- Promote electric bike memberships to attract casual riders.

Conclusion: The analysis indicates that casual riders and members exhibit different riding behaviors. By targeting casual riders with strategic promotions and membership incentives, Cyclistic can potentially increase long-term memberships and profitability.